

Discrete Mixing Operators from Boundary Sector Geometry

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Abstract

We develop a kinematic framework in which unitary mixing matrices arise as overlap operators between sector subspaces generated by boundary-localized states in a Hilbert space associated with a compact hyperbolic 3-manifold. Assuming a discrete assignment of sectors to boundary directions, we prove that the resulting mixing matrices are uniquely determined, up to sector-local rephasings, by the underlying boundary configuration. Our main result (Theorem 7.4) establishes that the set of admissible mixing matrices is discrete in $U(n)$ modulo $U(1)^n \times U(1)^n$, and that no continuous deformation of parameters can produce continuous drift in any rephasing-invariant mixing observable. Rephasing-invariant quantities depend only on the discrete boundary data and are stable under bounded deformations preserving the assignment rule. The construction is purely geometric and kinematic, requiring no dynamical input; it provides a model-independent structural constraint on theories in which fermions are associated with boundary directions in a compactified hyperbolic geometry.

1 Introduction

Mixing matrices, such as the CKM matrix in the quark sector and the PMNS matrix in the lepton sector, encode the mismatch between flavor and mass eigenstates in the Standard Model [5]. Despite their fundamental importance, their origin remains unexplained by the gauge structure alone. Many attempts to derive these matrices from first principles invoke discrete flavor symmetries, extra dimensions, or string-inspired constructions [1, 4]. In all such approaches, the mixing matrix ultimately measures the overlap between states associated with different sectors of the theory.

In this paper we take a kinematic, model-independent perspective. We show that if fermions are assigned to distinct boundary directions of a compact hyperbolic 3-manifold, the mixing matrices that arise from the overlap of the corresponding boundary-localized states are determined solely by the discrete boundary data. Moreover, these matrices form a discrete set modulo sector-local rephasings—a strong structural constraint that any realistic

model must satisfy. A corollary (Corollary 7.5) is that no continuous deformation of physical parameters can produce continuous drift in any rephasing-invariant mixing observable.

The construction is deliberately minimal: a Hilbert space of boundary data, a family of boundary-localized states, and a discrete assignment of fermions to boundary directions. No dynamics, no Lagrangian, and no ultraviolet completion are assumed. All results follow from elementary Hilbert space geometry and the discreteness of the boundary labels.

The paper is organized as follows. Section 2 establishes the geometric setup and the discreteness assumption. Section 3 introduces the boundary Hilbert space and localized states. Sector subspaces and preferred bases are defined in Section 4. Mixing matrices appear as overlap operators in Section 5. Rephasing invariance and its consequences are discussed in Section 6. Section 7 establishes stability under bounded deformations and the main discreteness theorem. We conclude in Section 8.

The geometric construction developed here provides the theoretical foundation for the companion results in [8, 9], where the CKM and PMNS mixing matrices are derived explicitly from the K - and N -factors of the Iwasawa decomposition of compact hyperbolic 3-manifold holonomy [8, 9].

2 Geometric Setup and the Discreteness Assumption

Let Γ be a torsion-free discrete group acting properly discontinuously and cocompactly on $\mathbb{H}^3$ by orientation-preserving isometries. The quotient

$$M = \mathbb{H}^3 / \Gamma$$

is a compact oriented hyperbolic 3-manifold. Its boundary at infinity is the sphere $\mathbb{S}^2 = \partial\mathbb{H}^3$.

Definition 2.1 (Boundary direction). A *boundary direction* is a point $\xi \in \mathbb{S}^2$. A group element $\gamma \in \Gamma$ is *associated with* ξ when the geodesic ray from a fixed basepoint $o \in \mathbb{H}^3$ through $\gamma \cdot o$ converges to ξ .

Assumption 2.2 (Discrete sector assignment). Each fermion flavor $f \in F$ is assigned a unique boundary direction $\xi(f) \in \mathbb{S}^2$ by a deterministic, discrete selection rule: the image $\Xi = \{\xi(f) : f \in F\}$ is finite, and the assignment is locally rigid.

Remark 2.3 (Geometric realization of the assignment). A concrete realization is as follows. Fix a basepoint $o \in \mathbb{H}^3$. For each fermion f , let $\gamma_f \in \Gamma$ be the unique minimal representative, i.e. the element minimizing $d(o, \gamma \cdot o)$ among all γ in the assignment class of f . (Uniqueness follows from the torsion-free cocompact action: proper discontinuity implies the orbit $\Gamma \cdot o$ is discrete, so for any class of elements the infimum $\inf_{\gamma} d(o, \gamma \cdot o)$ is realized; torsion-freeness then forces the minimizer to be unique. See [7] for the detailed proof in this setting.) The boundary direction $\xi(f)$ is then the ideal endpoint of the geodesic ray from o through $\gamma_f \cdot o$. The orbit $\Gamma \cdot o$ is discrete by proper discontinuity, so the set of ideal endpoints is discrete in $\mathbb{S}^2$; this gives local rigidity of the assignment.

3 Boundary Hilbert Space and Localized States

Let $\mathcal{H}$ be a separable complex Hilbert space carrying boundary data. A concrete realization is $L^2(\mathbb{S}^2)$, but any separable Hilbert space admitting a continuous family of normalized vectors $|\xi\rangle$ suffices.

Definition 3.1 (Boundary-localized state). For each $\xi \in \mathbb{S}^2$, fix a normalized vector $|\xi\rangle \in \mathcal{H}$ such that the map $\xi \mapsto |\xi\rangle$ is continuous and $\langle \xi | \xi \rangle = 1$ and $|\langle \xi | \zeta \rangle| < 1$ for $\xi \neq \zeta$.

The continuity condition ensures that nearby boundary directions give nearly parallel states; the normalization condition implies that distinct directions produce linearly independent vectors.

Remark 3.2 (Poisson kernel realization). An explicit family satisfying Definition 3.1 can be constructed using the Poisson kernel for $\mathbb{H}^3$. For $\xi \in \mathbb{S}^2$ and a basepoint $o \in \mathbb{H}^3$, set $|\xi\rangle$ to be the $L^2(\mathbb{S}^2)$ function proportional to the Poisson kernel $P(\cdot, \xi)$, normalized in $L^2(\mathbb{S}^2)$. The resulting family is continuous in ξ , satisfies $\langle \xi | \xi \rangle = 1$, and $|\langle \xi | \zeta \rangle| < 1$ for $\xi \neq \zeta$ [6]. In particular, distinct boundary directions produce linearly independent vectors, which is the property required by Definition 3.1.

4 Sector Subspaces and Preferred Bases

Let F be a finite set of fermion labels.

Definition 4.1 (Sector). A *sector* S is a finite ordered subset of F . The ordering is fixed once and for all (e.g., by increasing mass).

Definition 4.2 (Sector subspace). The *sector subspace* of S is

$$\mathcal{H}_S = \text{span}\{|\xi(f)\rangle : f \in S\} \subset \mathcal{H}.$$

Linear independence of the generating set implies $\dim \mathcal{H}_S = |S|$.

Definition 4.3 (Preferred orthonormal basis). The *preferred ONB* of $\mathcal{H}_S$ is the Gram–Schmidt orthonormalization of $(|\xi(f_1)\rangle, \dots, |\xi(f_{n_S})\rangle)$ in the fixed sector ordering, with all pivot coefficients real and positive. The resulting basis is denoted $\{|S, 1\rangle, \dots, |S, n\rangle\}$.

The choice of real positive pivot coefficients in the Gram–Schmidt process removes the $U(1)^n$ phase ambiguity in the definition of the preferred basis: every other ONB of $\mathcal{H}_S$ differs from the preferred one by a diagonal phase matrix $D_S \in U(1)^n$. This phase freedom is precisely the sector-local rephasing freedom reintroduced in Section 6 when comparing with physical observables. The preferred basis also depends on the sector ordering, but different orderings yield unitarily equivalent mixing matrices up to permutations, which can be absorbed into the rephasing freedom.

5 Mixing Matrices as Overlap Operators

Let S and T be two sectors of equal cardinality n .

Definition 5.1 (Mixing matrix). The *mixing matrix* U_{ST} is the $n \times n$ matrix with entries

$$(U_{ST})_{ji} = \langle T, j | S, i \rangle.$$

Proposition 5.2 (Unitarity). *If $\mathcal{H}_S = \mathcal{H}_T$, then $U_{ST} \in \text{U}(n)$. In general, U_{ST} is the matrix of the orthogonal projection $P_T|_{\mathcal{H}_S}$ in the preferred bases and is a partial isometry.*

Proof. When the subspaces coincide, the two preferred bases are orthonormal bases of the same space, so the change-of-basis matrix is unitary. In the general case, the operator $P_T|_{\mathcal{H}_S}$ is a partial isometry, and its matrix in the preferred bases is exactly U_{ST} . $\square$

Theorem 5.3 (Geometric dependence). *For fixed ordering conventions, U_{ST} is determined entirely by the finite boundary configuration*

$$\Xi_{ST} = \{\xi(f) : f \in S \cup T\}$$

and the inner product on $\mathcal{H}$.

Proof. All Gram–Schmidt coefficients are algebraic expressions in inner products $\langle \xi(f) | \xi(g) \rangle$, which depend only on the points in Ξ_{ST} and the fixed inner product. $\square$

6 Rephasing Invariance

In physical applications, unphysical phases can be removed by redefining the phases of fermion fields. Within a given sector, such redefinitions correspond to multiplying each basis vector $|S, i\rangle$ by an independent phase $e^{i\alpha_i}$, inducing

$$U_{ST} \mapsto D_T U_{ST} D_S^\dagger,$$

where $D_S, D_T \in \text{U}(1)^n$ are diagonal unitary matrices. Physical information is therefore contained in the equivalence class of U_{ST} in $\text{U}(n)/(\text{U}(1)^n \times \text{U}(1)^n)$.

Proposition 6.1 (Rephasing-invariant quantities). *The following are invariant under sector-local rephasings:*

- (i) *the moduli $|(U_{ST})_{ji}|^2$;*
- (ii) *the quartic cyclic products $J_{ji,j'i'} = (U_{ST})_{ji}(U_{ST})_{j'i'}\overline{(U_{ST})_{ji'}}\overline{(U_{ST})_{ji}}$;*
- (iii) *$\det(U_{ST})$ up to an overall phase.*

The quantities in (ii) generalize the Jarlskog invariant [3] to the present geometric setting. For $n = 3$ and the standard CKM or PMNS convention, the imaginary part $\text{Im}(J_{ji,j'i'})$ for any valid quartet (i, j, i', j') reduces to the standard Jarlskog invariant J_{CP} [3], which is basis-independent and rephasing-invariant by construction.

7 Stability and Discreteness of Mixing Classes

Definition 7.1 (Bounded deformation). Let $d_{\mathbb{S}^2}$ be a metric on $\mathbb{S}^2$ compatible with its topology. A *deformation of magnitude ε* replaces each $\xi(f)$ by a point $\xi'(f)$ with $d_{\mathbb{S}^2}(\xi(f), \xi'(f)) \leq \varepsilon$.

Theorem 7.2 (Stability). *Let $\delta_{\min} = \min_{f \neq f'} d_{\mathbb{S}^2}(\xi(f), \xi(f'))$ be the minimal pairwise separation of the original boundary directions, and set $\varepsilon_0 = \delta_{\min}/2$. For any deformation of magnitude $\varepsilon \leq \varepsilon_0$, the perturbed boundary configuration Ξ' determines the same preferred bases and the same mixing matrix U_{ST} .*

Proof. Here $d_{\mathbb{S}^2}$ denotes the round metric on $\mathbb{S}^2 = \partial\mathbb{H}^3$, and the assignment is the labeling $f \mapsto \xi(f)$ given by Definition 2.1 and Assumption 2.2. For $\varepsilon \leq \varepsilon_0$, each perturbed point $\xi'(f)$ remains closer to $\xi(f)$ than to any other $\xi(g)$ with $g \neq f$; hence the $\xi'(f)$ are distinct and each carries the same label as $\xi(f)$. Linear independence is therefore preserved. Since Gram–Schmidt coefficients depend continuously on inner products, and $\langle \xi'(f) | \xi'(g) \rangle$ varies continuously with ε , the preferred ONBs vary continuously. The discrete assignment rule (Assumption 2.2) is unchanged: each $\xi'(f)$ belongs to the same assignment class as $\xi(f)$. By Theorem 5.3, U_{ST} depends only on the configuration class, which is unchanged, so U_{ST} is unchanged. $\square$

Remark 7.3 (Geometric stability). In the geometric realization of Remark 2.1, the boundary direction $\xi(f)$ is the ideal endpoint of a geodesic ray, which depends continuously on the basepoint o . A small perturbation of o (or of the group element γ_f) moves $\xi(f)$ continuously on $\mathbb{S}^2$; for perturbations of magnitude $\varepsilon \leq \varepsilon_0$ this stays within the stability threshold and the assignment is unchanged.

Theorem 7.4 (Discreteness of mixing classes). *Under Assumption 2.2, the set of accessible mixing matrices*

$$\mathcal{M}_{ST} = \{U_{ST} : \Xi_{ST} \text{ admissible}\}$$

is discrete in $U(n)/(U(1)^n \times U(1)^n)$.

Proof. The set of admissible configurations Ξ_{ST} is finite. By Theorem 5.3, the map $\Xi_{ST} \mapsto U_{ST}$ is continuous, hence maps a finite set to a finite set. Quotienting by the rephasing action preserves finiteness, so $\mathcal{M}_{ST}$ is a finite—hence discrete—subset of the quotient. $\square$

Corollary 7.5 (No continuous drift). *For any continuous deformation of the boundary points $\xi(f)$ of magnitude $\varepsilon \leq \varepsilon_0$ (in the sense of Theorem 7.2), no continuous drift occurs in any rephasing-invariant mixing observable. For larger deformations that change the configuration class, the mixing matrix can change, but only by a discrete jump to another element of $\mathcal{M}_{ST}$.*

Proof. Rephasing-invariant observables are continuous functions on $U(n)/(U(1)^n \times U(1)^n)$. Their restriction to the discrete set $\mathcal{M}_{ST}$ takes only finitely many values; continuous drift is therefore impossible. $\square$

8 Conclusion

We have presented a purely kinematic construction of mixing matrices from boundary-localized states in a compact hyperbolic 3-manifold. The only inputs are a Hilbert space, a continuous family of normalized states $|\xi\rangle$, and a discrete assignment of fermions to boundary directions $\xi(f)$. From these, each pair of sectors gives rise to a unitary (or partially isometric) mixing matrix, uniquely determined up to sector-local rephasings. Our main theorem establishes that under the discreteness assumption the set of admissible mixing matrices is finite modulo rephasings—a strong structural constraint on any geometric theory of flavor mixing. A direct corollary is the absence of continuous drift in all rephasing-invariant observables.

This result is model-independent: it holds regardless of the dynamics that might generate the boundary assignment. In a companion paper we show how boundary directions can arise from unique minimal representatives in a cocompact group action [2, 6], linking the present kinematic framework to concrete arithmetic constructions. Future work could investigate whether the observed CKM and PMNS matrices belong to the discrete set for natural choices of boundary directions, or whether additional geometric structure is required [5].

Data Availability

This manuscript contains no research data. No data availability statement is required.

Conflict of Interest

The author declares no conflicts of interest.

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